

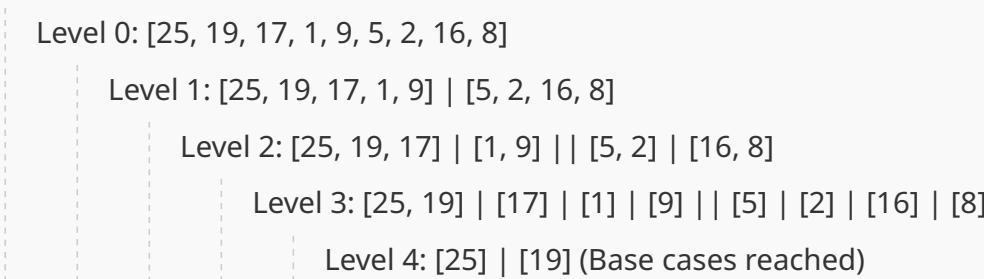
CS 3100/5100: Data Structures and Algorithms

Assignment 2 - Spring 2026 Solution Report

1. MergeSort Recursion Tree

Input Array: [25, 19, 17, 1, 9, 5, 2, 16, 8]

Divide Phase:



Merge Phase:

- Merge [25], [19] → [19, 25]
- Merge [19, 25], [17] → [17, 19, 25]
- Merge [1], [9] → [1, 9]
- Merge [17, 19, 25], [1, 9] → [1, 9, 17, 19, 25]
- Merge [5], [2] → [2, 5]
- Merge [16], [8] → [8, 16]
- Merge [2, 5], [8, 16] → [2, 5, 8, 16]
- **Final Merge:** [1, 9, 17, 19, 25] + [2, 5, 8, 16] → **[1, 2, 5, 8, 9, 16, 17, 19, 25]**

2. Minimum Spanning Tree (MST)

Algorithm Comparison & Results:

Step	Edge	Weight	Algorithm Notes
1	(4, 5)	3	Smallest edge in graph.

Step	Edge	Weight	Algorithm Notes
2	(s, 3)	5	Selected by Prim starting at s.
3	(2, 3)	8	Connects node 2.
4	(4, 6)	9	Cheapest connection for node 6.
5	(2, 4)	10	Connects top and bottom clusters.
6	(s, 7)	11	Connects node 7.
7	(4, t)	16	Direct path to sink node t.
8	(7, 8)	25	Mandatory to include node 8.

Total MST Weight: $3 + 5 + 8 + 9 + 10 + 11 + 16 + 25 = 87$

3. QuickSort Partitioning

Input: $S = \{6, 5, 3, 2, 7, 1, 4\}$ | **Pivot:** 4

j	A[j]	Action	i	Array State
-	-	Initial	-1	[6, 5, 3, 2, 7, 1, 4]
0	6	$6 > 4$, no swap	-1	[6, 5, 3, 2, 7, 1, 4]
1	5	$5 > 4$, no swap	-1	[6, 5, 3, 2, 7, 1, 4]
2	3	$3 \leq 4$, Swap A[0], A[2]	0	[3, 5, 6, 2, 7, 1, 4]
3	2	$2 \leq 4$, Swap A[1], A[3]	1	[3, 2, 6, 5, 7, 1, 4]
4	7	$7 > 4$, no swap	1	[3, 2, 6, 5, 7, 1, 4]
5	1	$1 \leq 4$, Swap A[2], A[5]	2	[3, 2, 1, 5, 7, 6, 4]
-	-	Final Swap A[i+1], Pivot	3	[3, 2, 1, 4, 7, 6, 5]

Result: Pivot 4 is placed at index 3.